



Sachin Kumar, Ph.D.

Process Development Leader

WORK EXPERIENCE

Sr. Staff Process Engineer

Prometheus Fuels, Santa Cruz, CA | July 2024 – July 2026

- Scale-up atmospheric CO₂ conversion technology into net-zero carbon fuels - Successfully demonstrating 1, 5 and **50 cell stacks for syngas production with S ratio of 2 or better.**
- Directed R&D lab operations and instrument calibration for critical analytical systems.
- Lead cross-functional engineering teams to develop, coat, and analyze next-gen electrodes.
- Automate material handling by programming and integrating a 6-axis robotic arm.
- Provide operational and technical data pipelines directly to the Founder/CEO.

Sr. Staff Process Development Engineer

Tesla Inc., Fremont, CA | Nov 2022 – Jun 2024

- Scaled cell manufacturing for the 4680-battery program - Process successfully validated and scaled at GFTX for **Model Y and Cybertruck battery packs.**
- Accelerated 4680 cell wetting by scaling dry battery electrode manufacturing from the electrolyte fill side, achieving a **50% reduction in E-Fill time to match wet electrode performance.**
- Developed proprietary multi-stage electrolyte filling hardware for rapid cell wetting.
- Co-designed laser-structured electrodes to maximize high-power cell output.
- Partnered with the Cell Engineering Manager and materials teams on core process goals.

Technical and R&D Manager

Microporous, Piney Flats, TN | May 2021 – Nov 2022

- Managed product development for specialized rubber and hybrid industrial battery separators.
- Accelerated commercialization of new formulations under harsh chemical conditions.
- Oversaw central R&D operations, standard testing procedures, and laboratory safety protocols.
- Represented the company on global technical committees to establish industry benchmarks.
- Reported to VP of Technology to align R&D initiatives with business goals.

Senior Scientist

Hollingsworth & Vose, West Groton, MA | July 2014 – Apr 2021

- Co-invented PowerAmp™ and Titan pasting papers for advanced automotive batteries, scaling product lines from initial concept into **\$31.5M+ in combined annual revenue.**
- Managed cross-functional projects developing EFB separators for Start-Stop vehicles.
- Created electrochemical testing methods adopted globally as a standard by the BCI.
- Evaluated competitor patents and shaped IP strategies to mitigate commercial risks.

Earlier Career History:

- **Senior Research Scientist** | Shamrock Energy Corporation | 2012 – 2014
- **Postdoctoral Research Scientist** | University of Wisconsin | 2010 – 2012
- **Visiting Assistant Professor** | Miami University | 2008 – 2009
- **Research/Teaching Associate** | Miami University | 2001 – 2008
- **Executive Engineer** | Ballarpur Industries Ltd. (BILT), India | 1999 – 2001

EDUCATION

2008 - Ph.D. Analytical Chemistry, Miami University

2004 - M.S. Paper Science & Engineering, Miami University

1999 - B.E. Pulp and Paper Technology, Indian Institute of Technology Roorkee

2021 – Mini MBA, Miami University

Selected Patents /Publications

- Full portfolio of 5 patents and 10 peer-reviewed papers available via [Google Scholar](#).
- **Patent (US11289771B2):** Battery components comprising fibers (Issued 2022).
- **Patent (US10014501B2):** Battery separators having a low apparent density (Issued 2018).
- **Publication:** "Electroreduction of O₂ on uniform arrays of Pt nanoparticles," J. Electroanal. Chem. (2013).
- **Publication:** "Electrooxidation of CO on uniform arrays of Au nanoparticles- Effect of particle size and interparticle spacing", Langmuir (2009).
- **Publication:** "Surface-enhanced Raman spectroscopy studies of 1,4-phenylene diisocyanide adsorption on Au and Pt-group transition metals", *Journal of Physical Chemistry B* (2006).

CONTACT

📞 920-750-0416

✉ kumar.sachin@gmail.com

🏠 Scotts Valley, CA, USA

🌐 www.linkedin.com/in/kumars1

SUMMARY

- 15+ year experience in R&D and Process Development
- Expert in Li-ion and Lead-acid battery, Supercapacitor and Carbon-capture technologies
- Experienced in developing concepts from lab-scale prototyping to high-volume manufacturing
- First-principles-based engineer that demands excellence in execution and creativity
- Identifies risks & deficiencies, and makes critical decisions in a fast pace environment
- Diverse background in building products and teams

TOOLS/SKILLS

- Process Development & Modeling
- Knowledge Management
- JIRA/Confluence
- Fundamental Engineering
- Validation Testing & Analysis
- Statistical Process Control
- Tolerance Analysis & Modeling
- Electron & Optical Microscopy
- Vibrational Spectroscopy
- Scanning Probe Microscopy
- Electrochemistry & Battery Testing
- Thin Film Deposition
- Failure Analysis
- Statistical Software and Data Analysis Tools

